

# CASPER MOLDRUP RYSGAARD

Computer Science Ph.D. in Algorithms and Data structures from Aarhus University

📍 Aarhus, Denmark  
in casper-rysgaard

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## MOTIVATION

## EXPERIENCE

### Ph.D. in Computer Science

#### Aarhus University

📅 August 2021 – July 2025

📍 Aarhus, Denmark

My Ph.D. was supervised by Gerth Stølting Brodal, and had a focus on Algorithms and Data structures, with a further focus on I/O efficiency and persistence, while covering several other topics. The Ph.D. taught me skills in project management, collaboration, reading and understanding previous techniques and solutions, creating new algorithmic solutions, writing, and presentation skills. Further, peer reviewing for conferences taught me to evaluate others' work. For a complete list of my results, see the later publications section.

### Teaching Assistant

#### Algorithms and Data Structures

📅 2018, 2019, 2020, 2021, 2022, 2023, 2024

#### Introduction to Programming with Scientific Applications (Python)

📅 2019, 2022, 2023

#### Programming Languages

📅 2020, 2021

Teaching has taught me explanation and presentation skills, allowing me to meet the students where they are, and grading assignments have furthered my peer reviewing skills, including feedback. From my later seniority I helped further at the weekly meetings of the teaching assistants, and at the final exam of the courses.

### Software Consultant

#### LBN Medical

📅 April 2016 – January 2017

📍 Aalborg, Denmark

LBN Medical buys and sells used medical equipment. I worked on their Microsoft Access database, building tools for the sales employees.

## EDUCATION

### Ph.D. in Computer Science

#### Aarhus University, Denmark

📅 August 2021 – September 2025

### MSc in Computer Science

#### Aarhus University, Denmark

📅 August 2020 – June 2023

Master's degree is obtained as first part of the Ph.D. The courses had focus on *algorithms, cryptology, logic and semantics* and *programming languages*.

🎓 Course Average: 12 (A+).

### BSc in Computer Science

#### Aarhus University, Denmark

📅 August 2017 – June 2020

Including 30 ects extra curriculum talent track. See later section for further details.

🎓 Course Average: 11.31 (A).

📄 Bachelor's Project: 12 (A+).

## VOLUNTEER WORK

### Chairman, TÅGEKAMMERET

📅 2022 – 2023

TÅGEKAMMERET is a party- and lecture organisation for students of Computer Science, Math, Physics, and more.

### Bartender, SMUKfest

📅 2022 –

SMUKfest is a festival in Skanderborg.

### Bartender, Fredagscaféen

📅 2018 – 2025

The Computer Science's friday bar.

### Kitchen Responsible, Regnecentralen

📅 2018 – 2025

A kitchen and social hub for students.

### Tutor, Mat/Fys-Tutorforening

📅 2018 – 2022, 2025

The tutors welcome the new students.

### Board Member, DSAU

📅 2019 – 2022

DSAU is a Computer Science student organization hosting social and skill learning events, in collaboration with various companies.

# COMPETITIVE PROGRAMMING

I highly enjoy competitive programming, and solving the style of tasks presented. In relation to this I have participated in the danish championship of competitive programming and other related competitions, mostly for fun. These are often hosted at the website *Kattis*. To view my user and ranking see: [open.kattis.com/users/casper-rysgaard](https://open.kattis.com/users/casper-rysgaard).

# LANGUAGES

English

Fluent

Danish

Native

# PUBLICATIONS

- A. Biniaz, A. Maheshwari, M. C. R. Merrild, J. S. B. Mitchell, S. Odak, V. Polishchuk, E. W. Robson, **C. M. Rysgaard**, J. K. R. Schou, T. C. Shermer, J. Spalding-Jamieson, R. Svenning, and D. W. Zheng, "Polynomial-time algorithms for contiguous art gallery and related problems," in *41st International Symposium on Computational Geometry, SoCG 2025, June 23-27, 2025, Kanazawa, Japan*, ser. LIPIcs, vol. 332, Schloss Dagstuhl - Leibniz-Zentrum für Informatik, 2025, 20:1–20:21. DOI: 10.4230/LIPICS.SOCG.2025.20.
- G. S. Brodal and **C. M. Rysgaard**, "Pure binary finger search trees," in *2025 Symposium on Simplicity in Algorithms, SOSA 2025, New Orleans, LA, USA, January 13-15, 2025*, SIAM, 2025, pp. 172–195. DOI: 10.1137/1.9781611978315.14.
- G. S. Brodal, **C. M. Rysgaard**, and R. Svenning, "Buffered partially-persistent external-memory search trees," in *33rd Annual European Symposium on Algorithms, ESA 2025, September 15-17, 2025, Warsaw, Poland*, ser. LIPIcs, vol. 351, Schloss Dagstuhl - Leibniz-Zentrum für Informatik, 2025, 82:1–82:18. DOI: 10.4230/LIPICS.ESA.2025.82.
- C. M. Rysgaard** and S. Wild, "Lazy b-trees," in *50th International Symposium on Mathematical Foundations of Computer Science, MFCS 2025, August 25-29, 2025, Warsaw, Poland*, ser. LIPIcs, vol. 345, Schloss Dagstuhl - Leibniz-Zentrum für Informatik, 2025, 87:1–87:19. DOI: 10.4230/LIPICS.MFCS.2025.87.
- G. S. Brodal, R. Fagerberg, and **C. M. Rysgaard**, "On Finding Longest Palindromic Subsequences Using Longest Common Subsequences," in *32nd Annual European Symposium on Algorithms (ESA 2024)*, ser. Leibniz International Proceedings in Informatics (LIPIcs), vol. 308, Dagstuhl, Germany: Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2024, 35:1–35:16, ISBN: 978-3-95977-338-6. DOI: 10.4230/LIPICS.ESA.2024.35.
- S. C. Sølvsten, **C. M. Rysgaard**, and J. van de Pol, "Random access on narrow decision diagrams in external memory," in *Model Checking Software - 30th International Symposium, SPIN 2024, Luxembourg City, Luxembourg, April 8-9, 2024, Proceedings*, ser. Lecture Notes in Computer Science, vol. 14624, Springer, 2024, pp. 137–145. DOI: 10.1007/978-3-031-66149-5\_7.
- G. S. Brodal, **C. M. Rysgaard**, J. K. R. Schou, and R. Svenning, "Space-efficient functional offline-partially-persistent trees with applications to planar point location," in *Algorithms and Data Structures - 18th International Symposium, WADS 2023, Montreal, QC, Canada, July 31 - August 2, 2023, Proceedings*, ser. Lecture Notes in Computer Science, vol. 14079, Springer, 2023, pp. 644–659. DOI: 10.1007/978-3-031-38906-1\_43.
- G. S. Brodal, **C. M. Rysgaard**, and R. Svenning, "External memory fully persistent search trees," in *Proceedings of the 55th Annual ACM Symposium on Theory of Computing, STOC 2023, Orlando, FL, USA, June 20-23, 2023*, ACM, 2023, pp. 1410–1423. DOI: 10.1145/3564246.3585140.

# EXTRA CURRICULUM

(Pre) Talent Track  
**Aarhus University, Denmark**  
📅 August 2017 – June 2020

📍 Aarhus, Denmark

Participation in extra 30 etcs credit curriculum during my bachelor at Aarhus university.

| Time        | Research area                  | Advisor                        | Project                                                                                              |
|-------------|--------------------------------|--------------------------------|------------------------------------------------------------------------------------------------------|
| 2017 Fall   | CS Challenge                   | Michael Schwartzbach           | 5 programming projects.                                                                              |
| 2018 Spring | Pre Talent Track               | Various advisors               | 5 introductory projects to the research groups.                                                      |
| 2018 Fall   | Logic and Semantics            | Aslan Askarov                  | Proof of dynamic non-interference for a small imperative language in Coq.                            |
| 2019 Spring | Algorithms and Data Structures | Gerth Stølting Brodal          | Selecting the $k$ smallest items from an infinite heap in $O(k)$ time using deterministic selection. |
| 2019 Fall   | Logic and Semantics            | Bas Spitters & Danil Annenkov  | Analysis of Liquidity Smart Contracts, and formal verification in Coq.                               |
| 2020 Spring | Cryptologi                     | Ivan Damgård & Sabine Oechsner | Formal verification of flooding network in Coq.                                                      |